

SYDNEY TECHNICAL HIGH SCHOOL



HIGHER SCHOOL CERTIFICATE ASSESSMENT TASK 1 MARCH 2014

Mathematics Extension 2

Name

Teacher

General Instructions

- Working Time - 70 min.
- Write using a blue or black pen.
- Board approved calculators may be used.
- A table of standard integrals is provided at the back of this paper.
- In Questions 6-9, show relevant mathematical reasoning and /or calculations.
- Begin each question on a fresh sheet of paper.

Total marks (53)

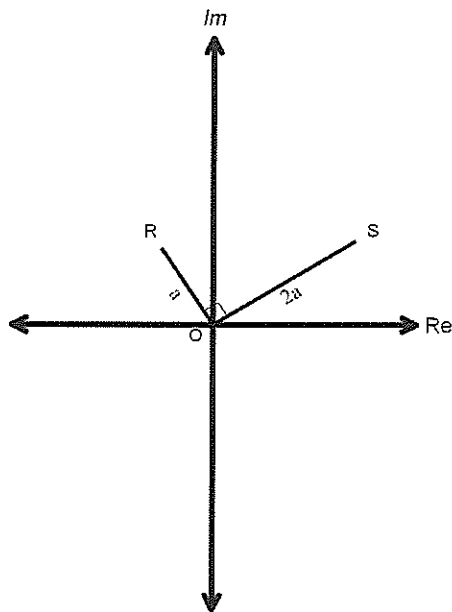
- Attempt Questions 1-9.
- All questions are of equal value.

Section 1

Multiple Choice (5 marks)

Use the multiple choice answer sheet for Question 1-5

1. In the Argand Diagram the locus of the point P representing the complex number z such that $|z - 1 + i| = 4$ is a circle. What are the centre and radius of this circle?
- (A) centre $(-1,1)$, radius 4
- (B) centre $(-1,1)$, radius 2
- (C) centre $(1,-1)$, radius 4
- (D) centre $(1,-1)$, radius 2
2. In the Argand Diagram below the points R and S represent the complex numbers w and z respectively, where $\angle ROQ = 90^\circ$. The distance OS is $2a$ units and the distance OR is a units. Which of the following is correct?



- (A) $w = 2iz$
- (B) $w = i\bar{w}$
- (C) $w = -\frac{iz}{2}$
- (D) $w = -\frac{z}{2i}$

3. Let $z = a + ib$ where $a \neq 0$ and $b \neq 0$.
Which of the following statements is false.

(A) $z - \bar{z} = 2bi$

(B) $|z|^2 = |z||\bar{z}|$

(C) $|z| + |\bar{z}| = |z + \bar{z}|$

(D) $\arg(z) + \arg(\bar{z}) = 0$

4. Which pair of equations gives the directrices of $4x^2 - 25y^2 = 100$

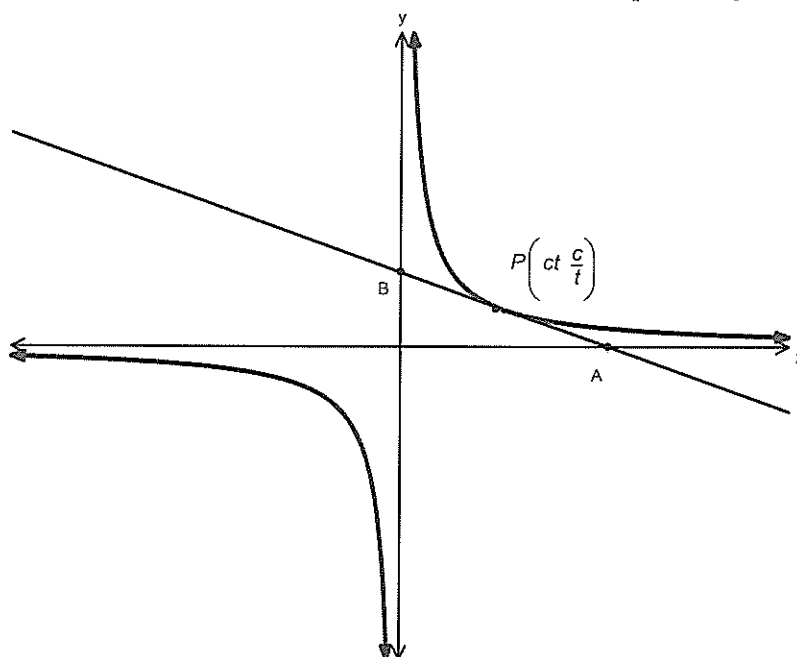
(A) $x = \pm \frac{25}{\sqrt{29}}$

(B) $x = \pm \frac{1}{\sqrt{29}}$

(C) $x = \pm \sqrt{29}$

(D) $x = \pm \frac{\sqrt{29}}{25}$

5. The equation of the tangent to the rectangular hyperbola $xy = c^2$ at $P\left(ct, \frac{c}{t}\right)$ is given by $x + t^2y = 2ct$. The tangent cuts the x and y axes at A and B respectively.



Which of the following statements is false?

- (A) P is the centre of the circle that passes through O , A and B .
- (B) The area of $\triangle AOB$ is $2c^2$ square units
- (C) The distance AB is $\sqrt{4c^2t^2 + \frac{4c^2}{t^2}}$.
- (D) $AP > BP$

Section II

Total Marks (48)

Attempt Questions 6 – 9.

Answer each question in your writing booklet.

In Questions 6-9, your responses should include relevant mathematical reasoning and/or calculations.

Question 6 (9 Marks)

Use a Separate Sheet of paper

- a) Given $A = 3 - 4i$ and $B = 5 + 3i$, express the following in the form $x + iy$ where x and y are real numbers.

i. $\frac{A}{B}$ 2

ii. $\sqrt{A}$ 2

- b) Show that the tangent to the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ at the point $P(x_0, y_0)$ has

equation: $\frac{xx_0}{a^2} + \frac{yy_0}{b^2} = 1$. 2

- c) On an Argand diagram, sketch the region where the inequalities

$2 \leq |z| \leq 5$ and $\arg \frac{\pi}{6} < \arg z < \frac{2\pi}{3}$ hold simultaneously 3

End of Question 6

Question 7 (9 Marks)

Use a Separate Sheet of paper

- a) i. Find the five fifth roots of $z^5 = 1$ and plot these on the Argand diagram. 2
- ii. Prove that $\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$. 3
- b) i. Expand $(\cos \theta + i \sin \theta)^3$ using Pascals triangle (or other) 1
- ii. Expand $(\cos \theta + i \sin \theta)^3$ using de Moivres theorem and hence show that
$$\sin 3\theta = 3 \sin \theta - 4 \sin^3 \theta$$
 3

End of Question 7**Question 8 (9 Marks)**

Use a Separate Sheet of paper

- a) A conic C has foci at $(4,0)$ and $(-4, 0)$ and has eccentricity, $e = \sqrt{2}$.
Find the equation of this conic 2
- b) i) Show that $P(2\sqrt{2} \cos \theta, 3\sqrt{2} \sin \theta)$ lies on the ellipse $\frac{x^2}{4} + \frac{y^2}{9} = 2$ 1
- ii) Show that the slope of the tangent at P is $\frac{-3 \cos \theta}{2 \sin \theta}$ 2
- iii) Find the equation of the normal to the ellipse at P 2
- iv) Find the value of θ to the nearest degree, if the normal passes through the point $(-2\sqrt{2}, 0)$ 2

End of Question 8

Question 9 (9 Marks)

Use a Separate Sheet of paper

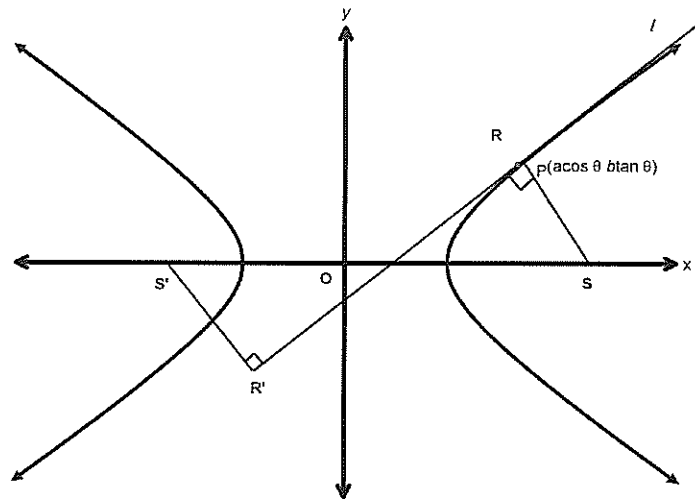
a) $z_1 = 1 + i\sqrt{3}$ and $z_2 = 1 - i$ are two complex numbers.

i. Express z_1 , z_2 and $\frac{z_1}{z_2}$ in modulus/argument form. 2

ii. Find the smallest positive integer n such that $\frac{(z_1)^n}{(z_2)^n}$ is imaginary.

For this value of n , write the value of $\frac{(z_1)^n}{(z_2)^n}$ in the form bi where b is a real number. 2

b) Let $P(a \cos \theta, b \tan \theta)$ be a point on the hyperbola $\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$ where $a > 0$ and $b > 0$ as shown in the diagram. The foci of the hyperbola are S and S' , l is the tangent to the point P .



The points R and R' lie on l so that SR and $S'R'$ are perpendicular to l .

The line l has equation $bx \sec \theta - ay \tan \theta - ab = 0$ 2

i. Show that $SR = \frac{ab(e \sec \theta - 1)}{\sqrt{a^2 \tan^2 \theta + b^2 \sec^2 \theta}}$ 2

ii. Show that $SR \times S'R' = b^2$ 3

End of Examination

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

Mathematics Extension 2

Assessment Task 1

March 2014

Multiple Choice

1. C

2. D

3. C

4. A

5. D

a ii) let $\sqrt{A} = x + iy$

$$\therefore A = x^2 - y^2 + 2xyi$$

$$x^2 - y^2 = 3 \dots\dots (1)$$

$$2xy = -4$$

$$(x^2 + y^2)^2 = (x^2 - y^2)^2 + 4x^2y^2$$
$$= 3^2 + 4^2$$

$$= 25$$

$$x^2 + y^2 = 5 \dots\dots (2)$$

Question 6

a i) $\frac{A}{B} = \frac{3-4i}{5+3i}$

$$= \frac{3-4i}{5+3i} \times \frac{5-3i}{5-3i}$$

$$= \frac{15-9i-20i-12}{25+9}$$

$$= \frac{3-29i}{34}$$

$$= \frac{3}{34} - \frac{29}{34}i$$

$$(1) + (2) = 2x^2 = 8 \quad (2) - (1) = 2y^2 = 2$$

$$x^2 = 4$$

$$x = \pm 2$$

$$y^2 = 1$$

$$y = \pm 1$$

Since $2xy = -4$

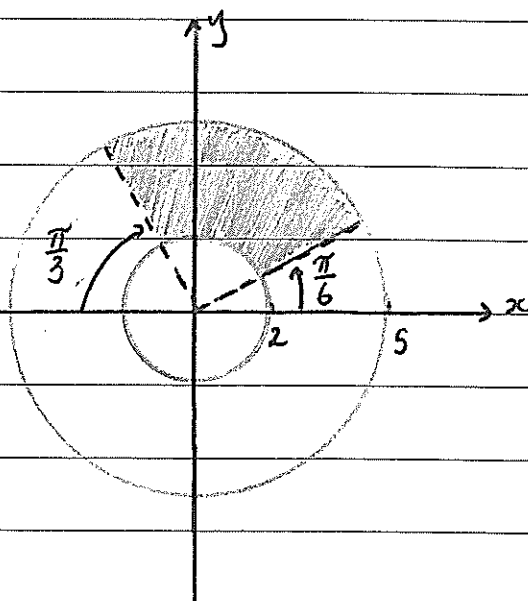
$$\sqrt{A} = \pm (2-i)$$

$$b) \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

$$\frac{2x}{a^2} + \frac{2y}{b^2} \frac{dy}{dx} = 0$$

$$\frac{2y}{b^2} \frac{dy}{dx} = -\frac{2x}{a^2}$$

$$\frac{dy}{dx} = -\frac{b^2 x}{a^2 y}$$



$$\text{At } (x_0, y_0) \quad y - y_1 = m(x - x_1)$$

$$m = -\frac{b^2 x}{a^2 y} \quad y - y_0 = -\frac{b^2 x_0}{a^2 y_0} (x - x_0)$$

$$a^2 y y_0 - a^2 y_0^2 = -b^2 x x_0 + b^2 x_0^2$$

$$b^2 x x_0 + a^2 y y_0 = a^2 y_0^2 + b^2 x_0^2$$

Dividing by $a^2 b^2$

$$\frac{x x_0}{a^2} + \frac{y y_0}{b^2} = \frac{x_0^2}{a^2} + \frac{y_0^2}{b^2}$$

$$\frac{x_0^2}{a^2} + \frac{y_0^2}{b^2} = 1$$

$$\therefore \frac{x x_0}{a^2} + \frac{y y_0}{b^2} = 1$$

Question 7

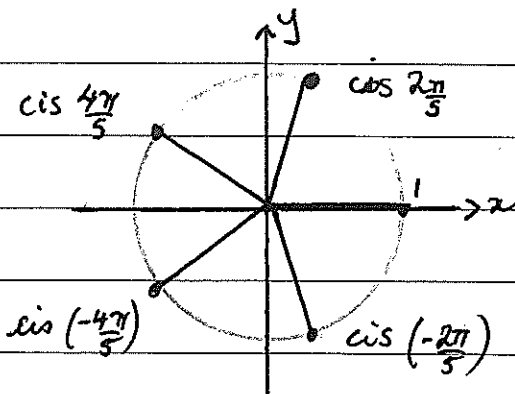
a i) $z_1 = 1$

$$z_2 = \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5}$$

$$z_3 = \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5}$$

$$z_4 = \cos \left(-\frac{4\pi}{5}\right) + i \sin \left(-\frac{4\pi}{5}\right)$$

$$z_5 = \cos \left(-\frac{2\pi}{5}\right) + i \sin \left(-\frac{2\pi}{5}\right)$$



a ii) Sum of root of $z^5 - 1 = 0$ is 0

$$z_1 + z_2 + z_3 + z_4 + z_5 = 0$$

$$1 + \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \frac{4\pi}{5} + i \sin \frac{4\pi}{5} + \cos \left(-\frac{4\pi}{5}\right) + i \sin \left(-\frac{4\pi}{5}\right) + \cos \left(-\frac{2\pi}{5}\right) + i \sin \left(-\frac{2\pi}{5}\right)$$

$$\text{since } \cos \frac{2\pi}{5} + i \sin \frac{2\pi}{5} + \cos \left(-\frac{2\pi}{5}\right) + i \sin \left(-\frac{2\pi}{5}\right) = 2 \cos \left(\frac{2\pi}{5}\right) = 0$$

$$1 + 2 \cos \frac{2\pi}{5} + 2 \cos \frac{4\pi}{5} = 0$$

$$2 \left(\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} \right) = -1$$

$$\cos \frac{2\pi}{5} + \cos \frac{4\pi}{5} = -\frac{1}{2}$$

b i) $(\cos \theta + i \sin \theta)^3$
$$= \cos^3 \theta + 3i \cos^2 \theta \sin \theta - 3 \cos \theta \sin^2 \theta - i \sin^3 \theta$$

b ii) $(\cos \theta + i \sin \theta)^3$
$$\cos 3\theta + i \sin 3\theta$$

Equating the imaginary parts
$$\sin 3\theta = 3 \cos^2 \theta \sin \theta - \sin^3 \theta$$

$$= 3(1 - \sin^2 \theta) \sin \theta - \sin^3 \theta$$

$$= 3 \sin \theta - 3 \sin^3 \theta - \sin^3 \theta$$

$$= 3 \sin \theta - 4 \sin^3 \theta$$

Question 8

$$a) \text{ Foci} = \pm ae$$

$$4 = a\sqrt{2}$$

$$a = \frac{4}{\sqrt{2}} = 2\sqrt{2}$$

$$b = 2\sqrt{2}$$

Rectangular hyperbola

$$\frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

$$\frac{x^2}{8} - \frac{y^2}{8} = 1$$

$$x^2 - y^2 = 8$$

$$b.ii) \frac{x^2}{4} + \frac{y^2}{9} = 2$$

$$P(2\sqrt{2} \cos \theta, 3\sqrt{2} \sin \theta)$$

$$L.H.S = \frac{(2\sqrt{2} \cos \theta)^2}{4} + \frac{(3\sqrt{2} \sin \theta)^2}{9}$$

$$\frac{8 \cos^2 \theta}{4} + \frac{18 \sin^2 \theta}{9}$$

$$2(\cos^2 \theta + \sin^2 \theta)$$

$$= 2$$

$$= R.H.S$$

$$b.iii) \frac{x^2}{4} + \frac{y^2}{9} = 2$$

$$\frac{x}{2} + \frac{2y}{9} \frac{dy}{dx} = 0$$

$$\frac{2\sqrt{2} \cos \theta}{2} + \frac{6\sqrt{2} \sin \theta}{9} \frac{dy}{dx} = 0$$

$$\frac{2 \sin \theta}{3} \frac{dy}{dx} = -\cos \theta$$

$$\frac{dy}{dx} = -\frac{3 \cos \theta}{2 \sin \theta}$$

$$b.iii) m_2 = \frac{+3 \sin \theta}{3 \cos \theta}$$

$$y - 3\sqrt{2} \sin \theta = \frac{2 \sin \theta}{3 \cos \theta} (x - 2\sqrt{2} \cos \theta)$$

$$3y \cos \theta - 9\sqrt{2} \sin \theta \cos \theta = 2x \sin \theta - 4\sqrt{2} \sin \theta \cos \theta$$

$$3y \cos \theta - 2x \sin \theta = 5\sqrt{2} \sin \theta \cos \theta$$

$$b.iv) P(-2\sqrt{2}, 0)$$

$$3y \cos \theta - 2x \sin \theta = 5\sqrt{2} \sin \theta \cos \theta$$

$$3 \times 0 \cos \theta - 2(-2\sqrt{2}) \sin \theta = 5\sqrt{2} \sin \theta \cos \theta$$

$$0 + 4\sqrt{2} \sin \theta = 5\sqrt{2} \sin \theta \cos \theta$$

3 solutions

$$\sin \theta = \underline{0^\circ}$$

$$4\sqrt{2} \sin \theta = 5\sqrt{2} \sin \theta \cos \theta$$

$$\cos \theta = \frac{4}{5}$$

$$\theta = \underline{37^\circ}, \underline{323^\circ}$$

Question 9

$$a_i \quad z_1 = 1 + i\sqrt{3}$$

$$z_2 = 1 - i$$

$$z_1 = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right)$$

$$= 2\left(\cos\frac{\pi}{3} + i\sin\frac{\pi}{3}\right)$$

$$z_2 = \sqrt{2}\left(\frac{1}{\sqrt{2}} - i\frac{1}{\sqrt{2}}\right)$$

$$= \sqrt{2}\left(\cos\left(-\frac{\pi}{4}\right) + i\sin\left(-\frac{\pi}{4}\right)\right)$$

$$\frac{z_1}{z_2} = \frac{2}{\sqrt{2}} \left(\cos\frac{7\pi}{12} + i\sin\frac{7\pi}{12}\right)$$

$$a_{ii}) \quad \arg\left(\frac{z_1^n}{z_2^n}\right) = \frac{7n\pi}{12}$$

$$\therefore \left(\frac{2}{\sqrt{2}}\right)^6 \left[\cos\left(\frac{7 \times 6\pi}{12}\right) + i\sin\left(\frac{7 \times 6\pi}{12}\right)\right]$$

$$\therefore \frac{z_1^n}{z_2^n} \text{ is imaginary if } \frac{7n\pi}{12} = \frac{m\pi}{2}$$

The smallest positive integer n is 6

$$\arg\left(\frac{z_1^6}{z_2^6}\right) = \frac{7\pi}{2} = 4\pi - \frac{\pi}{2} = -\frac{\pi}{2}$$

$$\left|\frac{z_1^6}{z_2^6}\right| = (\sqrt{2})^6 = 8$$

$$\frac{z_1^6}{z_2^6} = 8 \left[\cos\left(-\frac{\pi}{2}\right) + i\sin\left(-\frac{\pi}{2}\right)\right]$$

$$= -8i$$

b_i) Perpendicular distance

$$SR = \frac{|b \times a \sec \theta - 0 - ab|}{\sqrt{b^2 \sec^2 \theta + a^2 \tan^2 \theta}}$$

$$= \frac{ab|e \sec \theta - 1|}{\sqrt{a^2 \tan^2 \theta + b^2 \sec^2 \theta}}$$

$$\text{as } a > 0 \quad b > 0$$

$$= \frac{ab(e \sec \theta - 1)}{\sqrt{a^2 \tan^2 \theta + b^2 \sec^2 \theta}}$$

$$\text{as } e > 0 \quad \sec \theta > 1$$

$$b_{ii}) \quad S'R' = \frac{|b(-ae) \sec \theta - 0 - ab|}{\sqrt{b^2 \sec^2 \theta + a^2 \tan^2 \theta}}$$

$$= \frac{|-ab(e \sec \theta + 1)|}{\sqrt{a^2 \tan^2 \theta + b^2 \sec^2 \theta}}$$

$$\text{as } a > 0 \quad b > 0$$

$$= \frac{ab(e \sec \theta + 1)}{\sqrt{a^2 \tan^2 \theta + b^2 \sec^2 \theta}}$$

$$\therefore SR \times S'R'$$

$$= \frac{a^2 b^2 (e \sec \theta - 1)(e \sec \theta + 1)}{a^2 \tan^2 \theta + b^2 \sec^2 \theta}$$

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$$= \frac{a^2 b^2 (e^2 \sec^2 \theta - 1)}{a^2 \tan^2 \theta + b^2 \sec^2 \theta}$$

$$= \frac{a^2 b^2 (e^2 \sec^2 \theta - 1)}{a^2 \tan^2 \theta + a^2 (e^2 - 1) \sec^2 \theta}$$

$$b^2 = a^2 (e^2 - 1)$$

$$= \frac{a^2 b^2 (e^2 \sec^2 \theta - 1)}{a^2 [e^2 \sec^2 \theta - (\sec^2 \theta - \tan^2 \theta)]}$$

$$= \frac{b^2 (e^2 \sec^2 \theta - 1)}{(e^2 \sec^2 \theta - 1)}$$

$$= b^2$$

$$\therefore SR \times S'R' = b^2$$